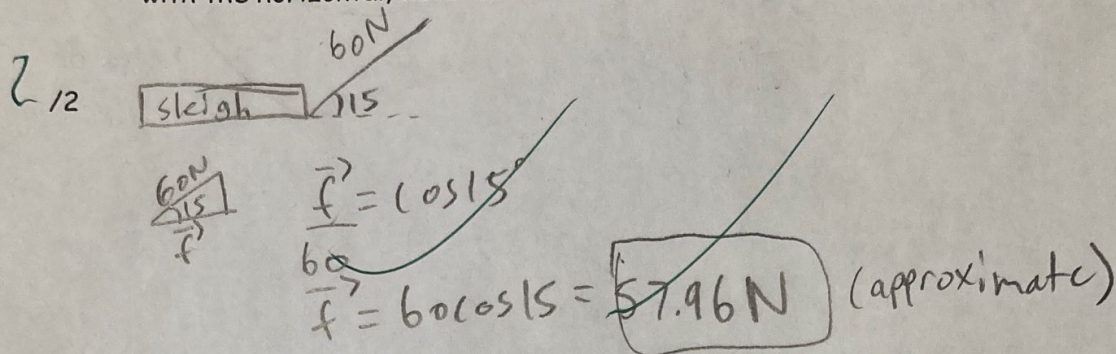


Name: _____

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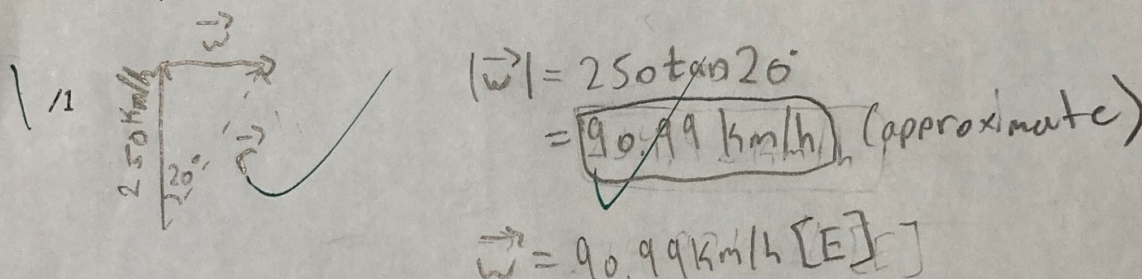
Total
14
/15

1. Paul pulls on a rope attached to his sleigh with a force of 60 N. If the rope makes an angle of 15° with the horizontal, determine the force that pulls the sleigh forward. Provide a diagram.



2. An airplane is travelling due north at an airspeed of 250 km/h when it encounters a wind from the west. The resultant ground velocity is in a direction N20E.

a) What is the rate at which the wind blows? Include a vector diagram as part of your solution.



b) How far has the plane travelled in 2 hours?

2/12

$v = \frac{d}{t}$

$v = |\vec{r}|$

$|\vec{r}|^2 = 250^2 + 90.99^2$

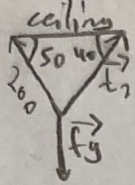
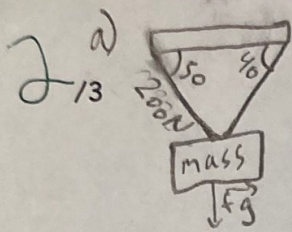
$|\vec{r}| = 266.09 \text{ km/h}$

$d = vt = 266.09(2) = 532.09 \text{ km (approximate)}$

$d = \text{distance}$

$t = \text{time}$

3. An object is hanging from two ropes. One rope has a tension of 200N and makes an angle of 50° with the ceiling. The other rope makes an angle of 40° with the ceiling. Find the tension in the second rope. Include a vector diagram in your solution. b) what is the mass of the object



$$\frac{|t_2|}{\sin 50} = \frac{200}{\sin 40}$$

$$|t_2| = 238.35 \text{ N (approximate)}$$

$$t_2 = 238.35 \text{ N } [40^\circ \text{ from ceiling}]$$

Must go tip to tail

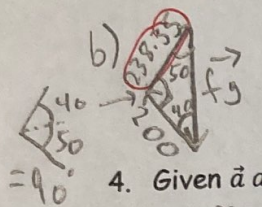
$$f_g = 9.8 \text{ m}$$

$$26.08 = 9.8 \text{ m}$$

$$m = 26.64 \text{ kg}$$

$$\cos 40 = \frac{200}{|f_g|}$$

$$|f_g| = \frac{200}{\cos 40} = 261.08 \text{ N (approximate)}$$

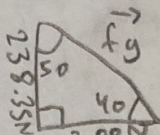


$$\vec{f}_g = \vec{t}_1 + \vec{t}_2$$

$$90^\circ - 40^\circ = 50^\circ$$

$$180^\circ - 90^\circ - 50^\circ = 40^\circ$$

rotate original triangle



4. Given $\vec{a}$ and $\vec{b}$ are unit vectors, the angle between them is 60° , calculate $(6\vec{a} + \vec{b}) \cdot (\vec{a} - 2\vec{b})$

$$(6\vec{a} + \vec{b}) \cdot (\vec{a} - 2\vec{b})$$

$$= 6\vec{a} \cdot \vec{a} - 12\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} - 2\vec{b} \cdot \vec{b}$$

$$= 6|\vec{a}|^2 - 12\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{b} - 2|\vec{b}|^2$$

$$= 6 - 11\vec{a} \cdot \vec{b} - 2$$

$$= 4 - 11\vec{a} \cdot \vec{b}$$

$$= 4 - 11|\vec{a}||\vec{b}|\cos 60$$

$$= 4 - 11(1)(1)(\frac{1}{2}) = 4 - 5.5 = -1.5$$

5. Which of the following sets of forces acting on an object could produce equilibrium? Include a sketch of how your choice(s) could be arranged to be in equilibrium.

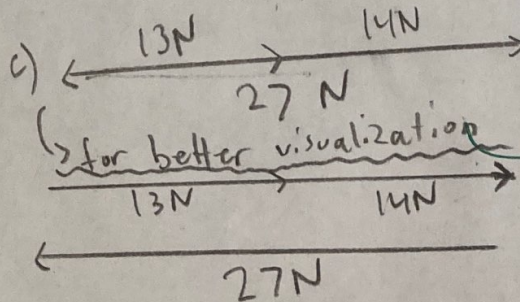
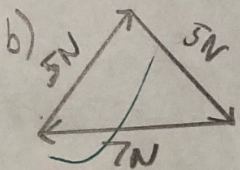
a) 5 N, 2 N, 13 N

b) 7 N, 5 N, 5 N

c) 13 N, 27 N, 14 N

d) 12 N, 26 N, 13 N

To have equilibrium, the largest force must be $\leq$ the other 2 forces combined. using this, only b) and c) can be arranged in equilibrium



(all of the vectors in part c have to be parallel to be in equilibrium)